

◆ APPENDIX A — MATHEMATICAL PROOFS (Final — Polished Version)

SEKHEM-TSDD Framework

A.1 Existence and Uniqueness

Consider:

$$\Psi_{k+1} = \Psi_k + dt \left(F(\Psi_k) + \sigma \sqrt{dt} \xi_k \right)$$

Assume:

- F is globally Lipschitz
- $\|F(\Psi)\| \leq C$
- Ω is compact

Then:

$$\|F(x) - F(y)\| \leq L \|x - y\|$$

By the Picard–Lindelöf theorem, the continuous-time system admits a unique solution.

For the Euler discretization, if:

$$dt < \frac{1}{L}$$

then the scheme is stable.

$\boxed{}$

`\text{A unique solution exists and is well-defined.}`
`}`

A.2 Boundedness

$$F =$$

- $\nabla \mathcal{E}$
- $L\Psi$
- + $\alpha(\text{PAC})\nabla \ln P$
- + $B(u)$

All terms are bounded, hence:

$$\|F\| \leq C$$

Recursion:

$$\|\Psi_{k+1}\| \leq \|\Psi_k\| + dt(C + \sigma\sqrt{dt}\|\xi_k\|)$$

Applying discrete Grönwall inequality:

$$\boxed{\|\Psi_k\| \leq C'}$$

A.3 Lyapunov Stability

$$\dot{L} = \dot{E}(\Psi) + KL(P - P_0)$$

Then:

$$\dot{L} \leq -c\|\Psi\|^2 + C\sigma^2$$

$\boxed{\text{The system is globally Lyapunov stable.}}$

A.4 Mean-Square Stability

$$\mathbb{E}\|\Psi_{k+1}\|^2 \leq (1 - c\,dt)\mathbb{E}\|\Psi_k\|^2 + C\sigma^2\,dt$$

Thus:

$$\mathbb{E}\|\Psi_k\|^2 \leq (1 - c\,dt)^k \|\Psi_0\|^2 + O(\sigma^2)$$

A.5 PAC Robustness

$$D = \mathrm{median}(R, C, P)$$

If one component diverges, the median remains bounded.

$\boxed{\text{Breakdown point} = 33\%}$

A.6 Decision Optimality

$$u_k = \arg\max_{u \in \mathcal{U}} D_u(k)$$

Since $\mathcal{U}$ is finite and D_u is bounded:

$\boxed{\text{An optimal decision exists.}}$

A.7 Closed-Loop Stability

$$\Psi_{k+1} = \Psi_k + dt F(\Psi_k, u_k)$$

With bounded feedback $u_k = \pi(\Psi_k)$, the dynamics remain Lipschitz.

$\boxed{\text{The closed-loop system remains stable.}}$

ADVANCED RESULTS

A.9 Delay Stability (Lyapunov–Krasovskii)

$$\dot{\Psi}(t) = F(\Psi(t)) - L\Psi(t-\tau)$$

Define:

$$V(t) = \|\Psi(t)\|^2 + \int_{t-\tau}^t \|\Psi(s)\|^2 ds$$

Under dissipativity and bounded coupling:

$$\dot{V}(t) \leq -\alpha V(t)$$

for sufficiently small delay:

$$\boxed{\tau < \tau_{\max}}$$

Thus:

$$\Psi(t) \rightarrow \Psi^*$$

A.10 $\mathcal{H}_{\infty}$ Robustness

$$\dot{e} = (A - BK)e + d(t)$$

Let:

$$V = e^T P e, \quad P \succ 0$$

If:

$$(A - BK)^T P + P(A - BK) \leq -Q$$

then:

$$\begin{aligned} \dot{V} &\leq \\ & - \lambda_{\min}(Q) \|e\|^2 \\ & + C \|d\|^2 \end{aligned}$$

Thus (in $\mathcal{L}_2$ sense):

$$\boxed{\|e\|_{\mathcal{L}_2} \leq \gamma \|d\|_{\mathcal{L}_2}}$$

A.11 Primal–Dual Convergence

$$\begin{aligned} \dot{\Psi} &= -\nabla_{\Psi} \mathcal{L}, \quad \text{quad} \\ \dot{\lambda} &= g(\Psi) \end{aligned}$$

Lyapunov:

$$V =$$

$$\begin{aligned} &\mathcal{E}(\Psi) - \mathcal{E}(\Psi^*) \\ &+ \frac{1}{2} \|\lambda - \lambda^*\|^2 \end{aligned}$$

Then:

$$\dot{V} \leq -\|\nabla_{\Psi} \mathcal{L}\|^2$$

By LaSalle:

$$\boxed{(\Psi, \lambda) \rightarrow (\Psi^*, \lambda^*)}$$

FINAL APPENDIX STATEMENT

\boxed{
\text{SEKHEM-TSDD is mathematically well-
posed, stable, robust, and constraint-satisfying.}
}

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